ACADEMY

Empowering For Professional Excellence

# Project Report

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**Course: Data Science**

**PITP – Batch 5, Phase 2 ,Group 2**

# Project Title: Student's Phone Addiction affects on Productivity.

# 1. INTRODUCTION:

The objective of this project is to analyze how smartphone usage and addiction affect students' productivity. Using a real-world dataset, we aim to identify key factors influencing productivity and build machine learning models to predict the productivity score.

# 2. Data Collection

- Dataset: Student Phone Addiction & Productivity Dataset

  Source: Kaggle

- Total Records: 10,000
- Features: 13 columns (numerical + categorical)

## Features:

- Daily Screen Time
- Study Hours
- Sleep Hours
- Social Media Usage
- Focus Score
- Addiction Level (categorical)
- Age
- notification per day
- Coffee per day

### Target Variable:

- Productivity Score (target)

# 3. Data Preprocessing

Data preprocessing was performed to clean and prepare the dataset:

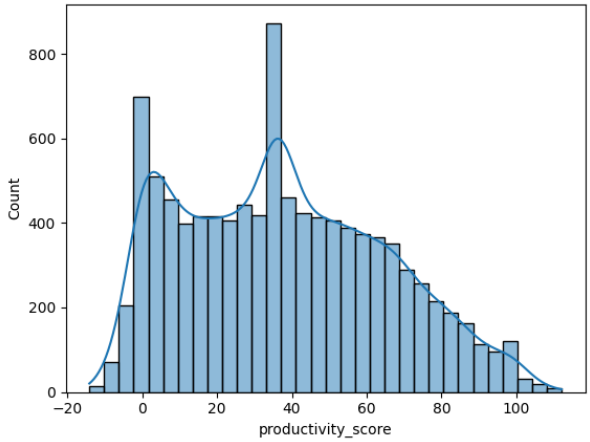
## Steps:

- **Check missing values**
- **Handled Missing Values:**
  - Numerical = filled using median
  - Categorical = filled using mode
- **Encoding:**
  - addiction_level converted into numeric using Label Encoding
- **Feature Removal:**
  - Dropped: age, coffee_per_day, notifications_per_day
- **Outlier Handling:**
  - focus_score capped at 100
- **Feature Scaling:**
  - Applied StandardScaler for KNN & Linear Regression

# 4.Exploratory Data Analysis (EDA)

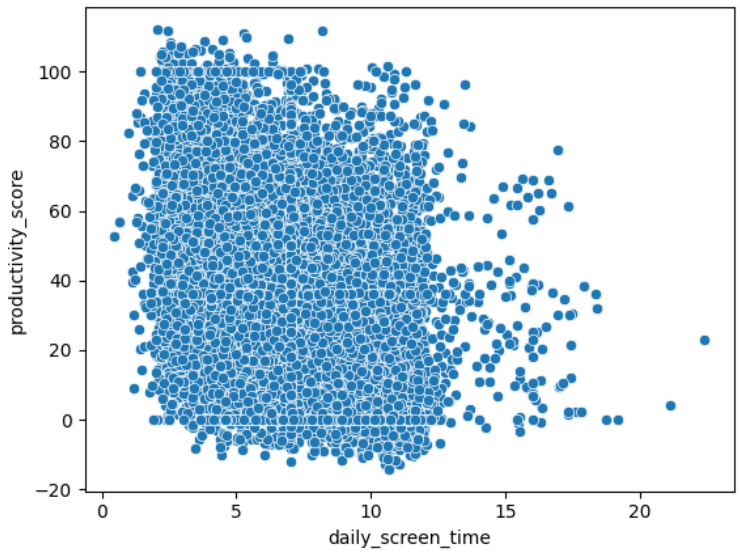
EDA is used to understand data patterns, relationships, and issues before building models

## Key Visualizations:




## Insights:

The productivity score follows an approximately normal distribution with most students in the mid-range. Few students have very low or very high productivity.




## Insights:

As daily screen time increases, productivity decreases

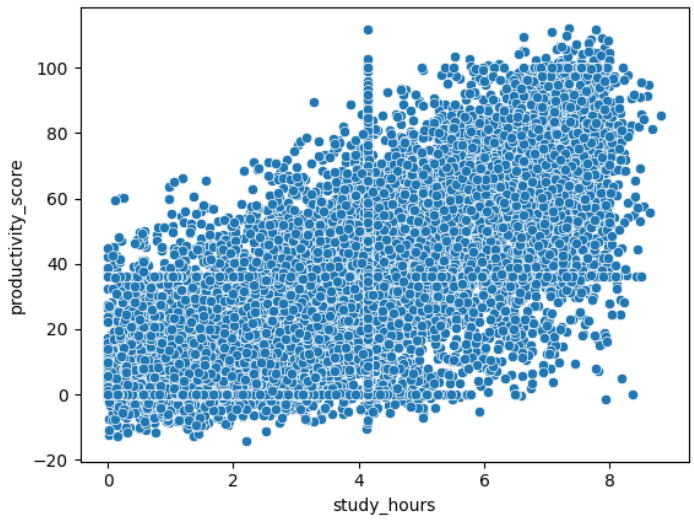
Some students:

- High screen time but still high productivity
- Low screen time but low productivity

Most data is between:

- Screen time: **3–10 hours**
- Productivity: **20–80**

This shows that phone usage impacts productivity, but it is not the only factor.

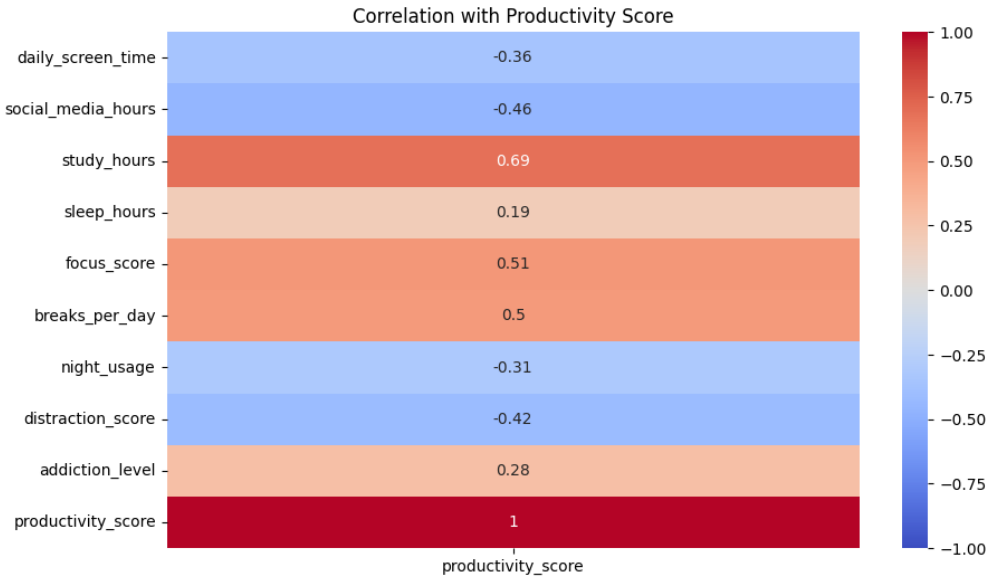



## Insights:

- Clear positive relationship
- As study hours increase,productivity increases

BUT:

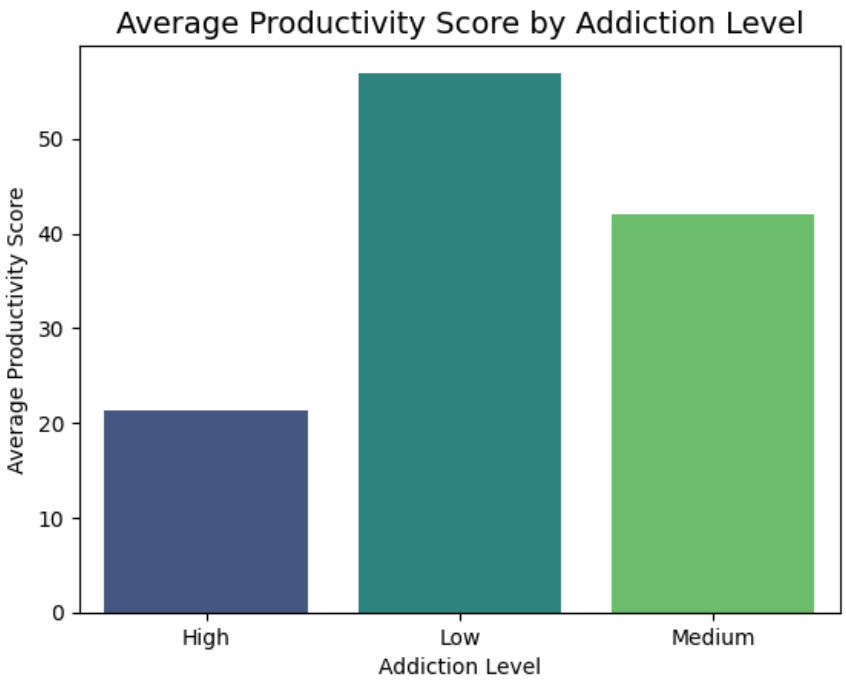
- Some students study more but still low productivity, other factors might affect them




## Insight:

- Study hours = 0.68
- Focus_score = 0.51

study habits and focus have the strongest positive correlation with productivity, while social media usage and distraction negatively impact it.




## Insights:

Students with high addiction levels have significantly lower productivity compared to low-addiction students.

# 5. Feature Engineering

New features help models learn better patterns

## 1. Study-to-Screen Ratio

**study_hours/daily_screen_time**

How much a student studies compared to how much they use their phone

## Logic:

- If a student:
    - studies more

- uses phone less
  ratio will be high
- If a student:
  - studies less
  - uses phone more
    ratio will be low

## 2. **Lifestyle Score**

$$(0.35\times study)+(0.30\times sleep)+(0.25\times focus)-(0.30\times social_media)$$

To create **ONE combined feature** that represents:

Overall student lifestyle quality

# Logic:

- **study_hours (0.35)**
  Most important = highest weight
- **sleep_hours (0.30)**
  Good sleep improves brain performance
- **focus_score (0.25)**
  Higher focus = better work

# 6. Model Building

Four regression models were implemented:

## 1. Random Forest Regressor:

- $R^2$ Score: **0.76**
- MAE: **10.09**
- Best performing model

## 2. Decision Tree

- $R^2$ Score: **0.72**
- MAE: **10.89**

## 3. K-Nearest Neighbors (KNN)

- $R^2$ Score: **0.73**

- MAE: **10.82**

### 4. Linear Regression

- $R^2$ Score: **0.74**
- MAE: **10.74**

# 7. Model Evaluation

## Metrics Used:

- **MAE (Mean Absolute Error):** Average prediction error
- **$R^2$ Score:** Model accuracy

## Comparison:

| Model | MAE | $R^2$ Score |
|---|---|---|
| Random Forest | 10.09 | **0.76** |
| Decision Tree | 10.89 | 0.72 |
| KNN | 10.82 | 0.73 |
| Linear Regression | 10.74 | 0.74 |

**Best Model:** Random Forest (highest accuracy, lowest error)

# 8. Tools & Technologies

- Python
- Pandas, NumPy
- Matplotlib, Seaborn
- Scikit-learn
- Google Colab

# Conclusion

- Productivity is not controlled by one factor
- It depends on a combination of habits (study, screen time, focus, lifestyle)